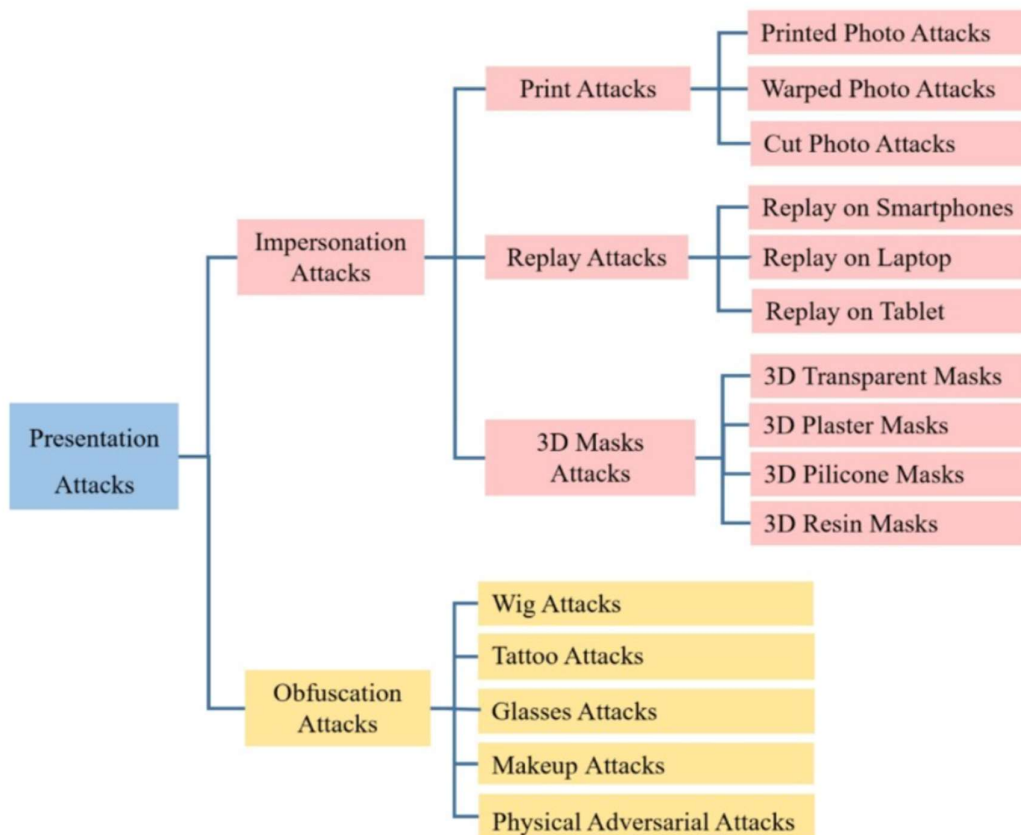
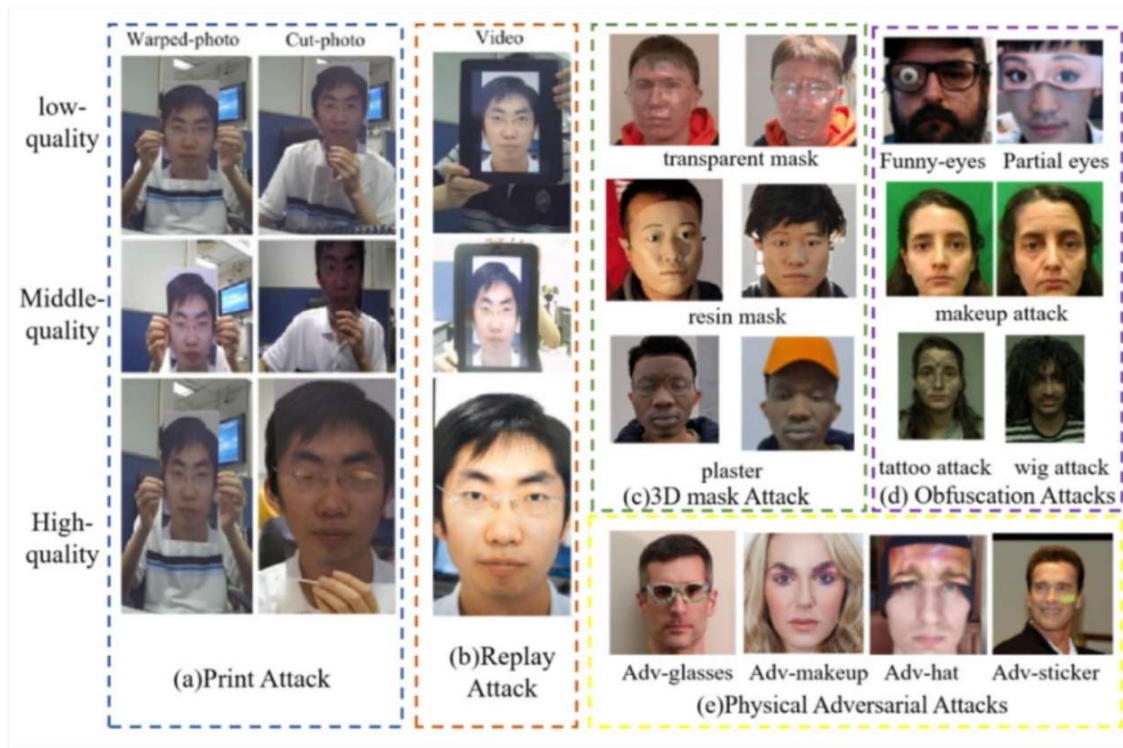


Face Anti-Spoofing for Robust Multi-Attack Liveness Detection



Title

A High-Performance Adaptive Fusion Network for Face Anti-Spoofing Detection

Journal Details

- **Journal:** Scientific Reports (Nature Portfolio)
- **Publisher:** Springer Nature
- **Year:** 2025

Introduction

The paper explains that face liveness detection is a biometric security technology used to distinguish genuine faces from spoof attacks such as printed photos, replay videos, and 3D masks. It is widely used in mobile phone unlocking, banking, attendance systems, and access control. As spoofing attacks become more sophisticated, there is a need for more robust deep learning-based face anti-spoofing methods.

Problem Identified

The authors state that existing face anti-spoofing methods have several limitations:

- Poor cross-domain generalization.
- Weak deep feature representation.
- Difficulty preserving facial details.
- Reduced performance under different lighting, devices, and attack types.
- Existing methods cannot efficiently learn common facial features across multiple domains.

Proposed Method

The authors propose a new model named:

DBC_ResNet18 (Dynamic Bottleneck Convolution ResNet-18)

The model is designed to improve cross-domain face anti-spoofing performance by enhancing facial feature extraction and reducing domain differences

The paper presents three major contributions.

Contribution 1

The authors use **ResNet-18** as the backbone network and introduce:

- Dynamic Convolution
- Bottleneck Attention Module (BAM)

These modules improve extraction of important facial regions and enhance feature representation across different domains.

Contribution 2

The authors combine:

- Central Difference Convolution (CDC)
- Adaptive Instance Normalization (AdaIN)

These techniques preserve facial details, improve feature representation, optimize style transfer, and increase the model's generalization capability.

Contribution 3

The proposed DBC_ResNet18 model reduces generalization error while maintaining high detection accuracy and efficiency on multiple benchmark datasets.

Technologies Used

The paper uses:

- Python
- PyTorch
- ResNet-18
- MTCNN
- PRNet
- Dynamic Convolution
- Bottleneck Attention Module (BAM)
- Adaptive Instance Normalization (AdaIN)
- Central Difference Convolution (CDC)
- Contrastive Learning
- Domain Adversarial Learning
- Gradient Reversal Layer (GRL)

Workflow

The workflow described in the paper is:

1. Data preprocessing
2. Face detection using MTCNN
3. Face cropping and resizing
4. Face depth map generation using PRNet
5. Feature extraction using ResNet-18
6. Dynamic convolution and Bottleneck Attention Module
7. Content feature extraction
8. Style feature extraction
9. Adaptive Instance Normalization
10. Central Difference Convolution
11. Contrastive learning
12. Domain adversarial learning
13. Binary classification (Real / Fake)

Data Preprocessing

The paper performs:

- Face detection using MTCNN.
- Face cropping.
- Resizing RGB images to **224 × 224**.
- Generating **32 × 32 depth maps** using PRNet.
- Real faces receive estimated depth maps, while fake faces use zero-valued depth maps

Backbone Network

The backbone network is ResNet-18.

It extracts common facial features from multiple source domains before further feature processing.

Dynamic Convolution

Instead of using fixed convolution kernels, the model dynamically adjusts convolution weights according to the input image.

This improves feature extraction and model adaptability across different datasets.

Bottleneck Attention Module (BAM)

The BAM combines:

- Channel Attention
- Spatial Attention

It focuses on important facial regions such as:

- Eyes
- Nose
- Mouth

This improves the extraction of discriminative facial features.

Adaptive Instance Normalization (AdaIN)

AdaIN is used for style transfer.

It aligns feature statistics from different domains while preserving important facial content. This improves cross-domain feature consistency.

Central Difference Convolution (CDC)

CDC enhances facial texture and edge information by incorporating pixel differences into convolution operations.

It improves detection of subtle spoofing artifacts.

Contrastive Learning

Contrastive learning is used to:

- Increase similarity among genuine face features.
- Separate spoof features.
- Enrich style features related to facial liveness.
- Reduce domain interference.

Domain Adversarial Learning

A domain discriminator and Gradient Reversal Layer are introduced.

The purpose is to reduce differences between multiple source domains and improve generalization on unseen datasets.

Loss Function

The overall loss combines:

- Classification Loss
- Adversarial Loss
- Contrastive Loss

These losses are optimized together using end-to-end training.

Datasets Used

The experiments are conducted on four public datasets:

- **OULU-NPU**
- **MSU-MFSD**
- **CASIA-FASD**
- **Replay-Attack**

Evaluation Metrics

The paper evaluates performance using:

- HTER (Half Total Error Rate)
- FAR (False Acceptance Rate)
- FRR (False Rejection Rate)
- AUC (Area Under Curve)
- ROC Curve
- Number of Parameters
- FLOPs
- Prediction Time

Experimental Results

The proposed DBC_ResNet18 model achieves:

- HTER as low as **0.06%** on some protocols.
- AUC up to **99.88%**.
- Single-image prediction time of **0.0388 seconds**.
- Better performance than several existing face anti-spoofing methods in cross-dataset evaluation

(Novelty)

Base Paper:

- Detects only **Live Face** or **Fake Face**.

Your Project:

- Detects whether the face is **Live** or **Fake**.
- If **Fake**, it identifies the **specific spoof attack type**:
 - Printed Photo
 - Replay Video
 - Mobile Screen Replay
 - 3D Mask
- Integrates the model into a **real-time authentication system** with a webcam and login decision.

Base Paper (They Used)	My Project (I Will Do)	Reason
RGB Image + Depth Map	Live RGB Webcam Video	Works with a normal webcam; no depth camera required.
MTCNN + PRNet	MediaPipe / YOLO Face	Faster face detection for real-time applications.
ResNet-18	EfficientNet-B0 / MobileNetV2	Better efficiency and accuracy for real-time deployment.
Dynamic Convolution + BAM	Built-in CNN Feature Extraction	Simpler model with lower computational cost.
AdaIN + CDC	CNN learns texture and reflection features	Reduces complexity while extracting spoof-related features.
Contrastive Learning + Domain Adversarial Learning (GRL)	Supervised CNN Training + Data Augmentation	Easier training and suitable for practical implementation.
Binary Classification (Live / Fake)	Multi-Class Classification	Detects the exact attack type.
Output: Live / Fake	Live, Printed Photo, Replay Video, Mobile Screen Replay, 3D Mask	Provides detailed spoof attack identification.
Research Model	Real-Time Authentication System	Allows login for live faces and blocks spoof attacks.
Dataset Evaluation	Dataset Evaluation + Live Webcam Testing	Validates both model accuracy and real-time performance.

Base Paper Datasets

Dataset	Purpose
OULU-NPU	Live vs. Print and Replay attacks
CASIA-FASD	Live, Print, Cut Photo, Video Replay
Replay-Attack	Live, Print, Replay attacks
MSU-MFSD	Live, Print, Mobile Replay attacks

Our dataset

CASIA-FASD

MSU-MFSD

Live Webcam

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Face Detection

(MediaPipe / YOLO Face)

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Face Cropping

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Image Preprocessing

(Resize 224×224, Normalize)

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Feature Extraction

(EfficientNet-B0 / MobileNetV2)

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Multi-Class Classification

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|— **Live Face**

|— **Printed Photo Attack**

|— **Replay Video Attack**

|— **Mobile Screen Replay Attack**

|— **3D Mask Attack***

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Authentication Module

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└─ Live Face → Login Allowed

└─ Spoof Attack → Access Denied

Tech Stack

- Python
- OpenCV
- MediaPipe (or YOLO Face)
- TensorFlow/Keras (EfficientNet-B0) **or** PyTorch
- NumPy
- Scikit-learn
- Flask/Streamlit (for UI)
- Webcam (OpenCV VideoCapture)

MediaPipe – Face Detection

OpenCV – Image Preprocessing

EfficientNet-B0 – Deep Learning Model

PyTorch – Model Training

CASIA-FASD + MSU-MFSD – Datasets

Scikit-learn – Evaluation Metrics

Abstract

Face recognition has become one of the most widely used biometric authentication techniques in applications such as smartphone unlocking, banking, attendance systems, and access control. However, conventional face recognition systems are vulnerable to presentation attacks, including printed photographs, replay videos, and mobile screen replay attacks, which can lead to unauthorized access. To address this challenge, this project proposes a **Face Anti-Spoofing for Robust Multi-Attack Liveness Detection** system using deep learning.

The proposed system captures live facial images through a webcam and detects the face using **MediaPipe Face Detection**. The detected face is preprocessed through cropping, resizing, and normalization before being passed to an **EfficientNet-B0** deep learning model for feature extraction. Unlike conventional binary face anti-spoofing methods that classify only **Live** or **Fake**, the proposed system performs **multi-class classification** to identify specific spoof attack types, including **Printed Photo Attack**, **Replay Video Attack**, and **Mobile Screen Replay Attack**, in addition to genuine live faces. Based on the prediction, the authentication module grants or denies access to the user.

The model is trained and evaluated using the **CASIA-FASD** and **MSU-MFSD** benchmark face anti-spoofing datasets. Performance is assessed using standard evaluation metrics such as **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**. The proposed system aims to improve the practicality of face anti-spoofing by combining real-time detection with detailed spoof attack identification, making it suitable for secure biometric authentication applications.